BullSO 2026 Invitational

Codebusters B

Bull Science Olympiad at the University of South Florida

Saturday, January 24th, 2026

Team Name:_____

Participant Names:_____

Total Score:_____/xxx **Rank:**_____

Science Olympiad — BullSO2026 Div B

Names of participants: (Please print neatly)

_____

_____

_____

School Name: _____ Team Number: **B**

Check your team: ☐ Varsity ☐ JV1 ☐ JV2 ☐ JV3

Warning: Do not open this packet until given permission to do so.

Note: There are useful notes after this page.

Scoring: Time to solve first problem: _____ (use to calculate Bonus below)

Question	Value	Incorrect letters	Deduction	Score
Timed	295			
1★	123			★
2	209			
3	447			
4	243			
5	206			
6	245			
7	532			
8★	205			★
9	259			
10	145			
11	245			
12	237			
13	256			
14	255		≡	
15	218			
16★	332			★
17	523			
18	105			
19	273			
20	321			
21	216			
22	227			
Bonuses	**Timed:**	**Special Bonus: ★ = 150; ★★ = 400; ★★★ = 750**		
Final Score				

Timed Question **[295 points]** A quote has been encoded using the Aristocrat Cipher for you to decode. When you have solved it, raise your hand so that the time can be recorded and the solution checked. When you have solved it, raise your hand so that the time can be recorded and the solution checked.

PZ NLTHHI YTD'Z GL ZKTZ KTNW ZB CTXL T QBNZI-QBVN

ITNW QPLHW EBTH. IBV WLOLNFL ZB ELZ OHPCLW BVZ TQZLN

CPOOPDE ZKTZ.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency		7	3	2	3	1	1	5	4		3	10		7	4	5	4			10		3	5	1	1	12
Replacement																										

★(Special Bonus Question) 1) **[123 points]** Your school's intercom crackled to life with static… then spelled out this Affine message by Benjamin Franklin with ***a*=21** and ***b*=10**. What does it say?

E	Q	H	H	V	S	X	Q	W	Y	F	Q	T	T	Q	D	T	B	K	X	E	Q	H	H	Y	K	W	V	.
																												.

2) **[209 points]** You open a forgotten email draft saved in the school's computer system. It holds a quote encoded in a Complete Columnar. The quote has **BUBBLES** somewhere in it. What does it say?

ORSLO HOAIR XFSBW MIAIO TXHEE UESDN WAXHU NOTEA

YONXI TBERR SSPUG TWLOG CNOPW XGBIF EPCLH IXUEM

DMACL TNX

Answer: _____

3) **[447 points]** A vending machine just dispensed a candy wrapper with this Fractionated Morse code by Goethe. You are told that the quote has **WHAT** in it corresponding to the encoded text **A Q I Y P**, meaning **A** = ●—; **I** = ●●×; **P** = –××; **Q** = ×●●; **Y** = ●–×. What does it say?

Y V S U X I B Q E I J W F B

X M L S B G U E A Q I Y P F B

Z D S S B J I B I Q U X F Q Q E

F B S S C

Replacement																									
Morse fraction	●	●	●	●	●	●	●	●	●	–	–	–	–	–	–	–	–	–	×	×	×	×	×	×	×	×
	●	●	●	–	–	–	×	×	×	●	●	●	–	–	–	×	×	×	●	●	●	–	–	–	×	×
	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–

4) **[243 points]** A fragment of text is discovered in a time capsule from 1965. It's a quote encrypted as a Checkerboard Cipher with a polybius key of **CUMBERLAND**. What does it say?

HR EN EP CT CP HI HN AP HN HR

PN CN CT CR HN HT CT HP AN AT

AP AP HT CN HI CP ET PR HI HP HT AN CN CR AT EN AT CI CR AN

AT CN CT HR EN PP CT HT EP AP HP PR HI HP HT AN

5) **[206 points]** A riddle is discovered taped under a chemistry lab bench. It's a quote by Francois de La Rochefoucauld, encrypted as an Aristocrat. What does it say?

U STJY CTAYFE AK SOY NPKS BTYXAPJK PC UDD

BPKKYKKAPFK UFE SOY PFY QY SUMY SOY DYUKS SOPJZOS

ULPJS UXRJATAFZ.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	6	2	2	3	2	5				5	9	1	1	1	5	8	1	1	10	4	7			2	11	2
Replacement																										

6) **[245 points]** While debugging a program, you find this hidden in the comments. It's a quote by John Marshall, encrypted with a Complete Columnar. The quote has **LISTEN** somewhere in it. What does it say?

LWARE CINFE ALTEI WAOUI INOEI ESFAO CALAL XSLPU

NMANU SKXON SEMFN ONCTL TLOLS MTDET WX

Answer: _____

7) **[532 points]** Decrypt the following cipher text that has been encoded using the Patristocrat Cipher using a K1 alphabet. What does it say?

QKIFA DNPBQ KYHNC JBWIC QAUPI HNYYI CWIWZ PNWXH

QEGFE IWNCY UGPNS EBCDK BEQNP USKBH KYNEQ AUPIJ

AIHQE GNPIW NYFCW YBAWH NCJXE BNC

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	5	7	8	2	7	3	3	6	9	3	5			12		7	7		2		4		8	2	7	1
Replacement																										

★(Special Bonus Question) 8) **[205 points]** Someone left this coded message in the school lab. You are told it's a quote by Marcus Aurelius encoded with a Nihilist Substitution with a keyword of **JULIEN** and polybius key of **ABOLISHMENT**. What does it say?

29 58 35 36 39 46 40 58 45 37

39 50 50 69 28 36 48 37 60 76

	1	2	3	4	5
1					
2					
3					
4					
5					

46 37 35 50 50 69 25 40 57 57 37 56 39 50 48 40 36 70 25 46

69 69 39 66 47 39 38 40 50 67 45

9) **[259 points]** Found inside a dissected frog's stomach (plastic, don't worry): this Aristocrat note. You are told that the cipher **starts with the word "The"**. What does it say?

ZSY MPKJIDZUFUZA EIPI QRPTYJ CY ROKIZ JUVZPRDZUKTV.

U UCCYJURZYGA VIOVDPUOYJ ZK ZSPYY CKPY TYQVGYZZYPV.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	2		4	3	1	1	2		5	5	5		1		3	8	2	4	2	3	8	6			12	11
Replacement																										

10) **[145 points]** You uncover this encoded note tucked behind a microscope slide. It's a quote by Lily Tomlin, written in a Porta using a keyword of **ADVISOR**. What does it say?

V	O	V	F	W		E	K	J	O	D		C	N	X	L	B		P	O	B	F	T		Z	O	A	N	H		X	A	L	V	E

R	K	L	S	Y		U	O	L	V	X		Y	Z	A	N	E		A	N	L	H	R		Q	S	W	R	W

11) **[245 points]** Someone left this encrypted quote scribbled on the chalkboard before class began. It's a quote, written as an Aristocrat. What does it say?

DPRZ DEAX VR DE DWFOD DPR CWEBROO. DPR CWEBROO IEK

PMO VR HMAMIBUIL OCEEIO QEW IE BARMW WRMOEI.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	3	4	3	7	10	1		1	6		1	1	5		9	4	1	9			1	2	6	1		1
Replacement																										

12) **[237 points]** Solve this Baconian by Byron Pulsifer. You are told that the deciphered text starts with **Fate.**

BIRDS ABIDE PHASE CARLA BRIDE NICKY A WALK LOYAL

OF ONE JONES OCCUR OBAMA ABOUT AGILE ARNIE ADIOS

REIGN ADELE JORGE ABBOT A ROAD ATOMS BONES ANNIE

AHOLD CARLA ANDRE NICKY ALPHA SHEEP

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Replacement																										

13) **[256 points]** A laser-etched message on the inside of a geodes crystal reads: solve this Aristocrat version of Ralph Emerson's wisdom. What does it say?

IYAMH MYA HJAR PJD VAA HJMH VSWYWHQMT WV VHYDXIAY

HJMX MXR UMHAYWMT GDYLA, HJMH HJDQIJHV YQTA HJA

PDYTN.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	10			5			1	13	3	8		1	9	1		2	3	2	1	4	1	5	4	3	9	
Replacement																										

14) **[255 points]** Archaeologists unearth a stone tablet inscribed with a message. It's a quote encoded as a Cryptarithm. What do the values `123423506 7586` decode to?

Values to decode for solution

1	2	3	4	2	3	5	0	6		7	5	8	6

Cryptarithm formula

```
  E F F O R T
+ P R O J E C T
---------------
A T T E M P T
```

	0	1	2	3	4	5	6	7	8	9
A										
C										
E										
F										
J										
M										
O										
P										
R										
T										

15) **[218 points]** Scrawled in dry-erase marker on the whiteboard after hours: this Aristocrat encoded quote by Helen Keller. What does it say?

QPL GJYQ OLEZQFMZV QPFBIY FB QPL DJKVC HEBBJQ OL

YLLB JK LXLB QJZHPLC. QPLT GZYQ OL MLVQ DFQP QPL

PLEKQ.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency		6	2	2	3	4	2	2	1	5	3	14	2		3	8	13			1		3		1	4	4
Replacement																										

★(Special Bonus Question) 16) **[332 points]** The message below was encoded into the DNA sequence of a synthetic organism. It translates to a quote by Buddha, encrypted using a Checkerboard Cipher with a polybius key of **CONSTABLE**. What does it say?

AR AS EH TR SS SH GE TO TE SR

TR TR SE AO TR GS AR GR TS SS

TO TR SH AS TS SR AR SO AH TO TR AS TO SH EH TR GR SS TO TR

TS GR TO ER SH GE GR EH AO SH TO TR TO AR AS TR EH SH GE TO

TE SS AO TS SO AH TR

17) **[523 points]** Hidden in the source code of a retro video game, this Patristocrat puzzle reveals a quote by Nelson Mandela using a K1 alphabet. You are told that the cipher contains **`the word "altered"`**. What does it say?

RYKQK CXDZR YCDEF CBKQK RSQDC DERZH OFHAK RYHRQ

KNHCD XSDAY HDEKJ RZLCD JRYKU HGXCD UYCAY GZSGZ

SQXKF LYHTK HFRKQ KJ

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	3	1	8	9	3	4	3	8		3	12	2		1	1		6	9	4	1	2			4	8	5
Replacement																										

18) **[105 points]** Your solar-powered calculator started flashing this Caesar sequence—it's Brian Tracy's words. What does it say?

G	W	C		P	I	D	M		E	Q	B	P	Q	V		G	W	C	,		Z	Q	O	P	B		V	W	E	,
																			,											,

M	D	M	Z	G	B	P	Q	V	O		G	W	C		V	M	M	L		B	W		L	M	I	T		E	Q	B	P

E	P	I	B	M	D	M	Z		B	P	M		E	W	Z	T	L		K	I	V		B	P	Z	W	E		I	B		G	W	C	.
																																				.

19) **[273 points]** You intercepted this transmission from a rogue weather balloon. Decode the Aristocrat version of Byron Pulsifer's famous words. What does it say?

CA UR CBAXMBCNXJ DQZ FGQZ AQJS CDFRV D NRO VRYAQZV

YAKHDERZ CA CBR CGKRJRVV BXEC YDXVRZ US AQR EXZR

MRVCXER.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	7	4	9	5	4	2	2	1		3	2		2	2	1		5	13	2		2	7		6	3	6
Replacement																										

20) **[321 points]** A message flashes briefly on a malfunctioning computer screen. It turns out to be a quote written in an Aristocrat using a K1 alphabet. What does it say?

IS AICMNQCNEB XFNFENFX ANMFAA. NBF ANMFAA QCA IVANWS

YMVI MFEFZPZJU EVJANCJN CWFMNA CDVON IS ANMFAA.

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	14	2	6	1	4	10			5	3			7	13	1	1	2		3		1	4	2	2	1	2
Replacement																										

21) **[216 points]** Your graphing calculator won't stop displaying this Aristocrat puzzle by Thomas Carlyle using a K1 alphabet. What does it say?

PE MOAKGH WOM SOAHI JSOMNH; IHVQXW CLVV SH ZKLA SLI

XVQ SXWH ALVV LA SOI OJAKOVVE ROVVHM OGXKMQ SLI HOGI.

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	6		1		2		3	8	6	2	4	6	5	1	10	1	3	1	7			10	3	4		1
Replacement																										

22) **[227 points]** A note falls out of a borrowed library book. The message is a quote by Anne Frank, written as an Aristocrat Cipher. What does it say?

YJS SJOWQIUKF NB NA BYVB OJZJWR OQQW SVNB V ANOMFQ

DJDQOB ZQUJIQ ABVIBNOM BJ NDXIJGQ BYQ SJIFW.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	3	9		3		3	1		5	9	1		2	6	6		9	1	4		2	4	4	1	3	2
Replacement																										

BullSO, University of South Florida - Repaired practice edition

Division B | Codebusters | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

Name: _____ Date: _____

repair-case. (6 points) Authored written substitute for unavailable source material: Solve this case and explain your reasoning: Puzzle A offers 100 points with a 0.9 estimated solve probability in 3 minutes; B offers 200 with 0.3 probability in 6 minutes. Compare expected points per minute as a planning aid.

Answer: _____
_____
_____